Explore Relationships Between Two Variables

Build a table of values and find the rule that connects the input to the output.

Name _____ **Date** _____

Tonight's spark

A faucet at the rec center fills a tub with 11 gallons each minute. You make a table for 1, 2 and 4 minutes. How many gallons are in the tub after 4 minutes?

Guess first: is it more than 40 gallons, or less?

Minutes (input)	Gallons (output)
1	11
2	22
4	?

+11

Our class model: the input-output table rule

Try it

Work it out on paper. Write one answer on each line.

1 **In a table of values, where does the input go?**

A. Column 2 **B.** Column 1 **C.** The title **D.** The bottom row

My answer _____

2 **A table shows x = 1, 2, 3, 4 and y = 5, 10, 15, 20. What is y when x = 3?**

A. 15 **B.** 10 **C.** 20 **D.** 3

My answer _____

3 **A table shows x = 1, 2, 3 and y = 3, 6, 9. How much does y change each time x goes up by 1?**

A. 1 **B.** 2 **C.** 3 **D.** 9

My answer _____

4 **A table shows x = 1, 2, 3 and y = 7, 9, 11. What is y when x = 4?**

A. 11 **B.** 12 **C.** 14 **D.** 13

My answer _____

5 **A table shows x = 1, 2, 3 and y = 2, 4, 6. Which rule connects y to x?**

A. y = x + 2 **B.** y = 2x **C.** y = x – 2 **D.** y = 3x

My answer _____

6 **For the rule y = x + 4, what is y when x = 5?**

A. 20 **B.** 1 **C.** 9 **D.** 45

My answer _____

Talk about it

Which column is the input and which is the output? What is the rule that connects them?

Home mission

Take a small cup and a bowl. Pour 2 cups at a time. Make a table for 1, 2 and 3 pours. Can your student say which column is the input and tell you the rule?

If we get stuck, ask

- Which quantity could you choose, and which one follows from it?
- Say the sentence: "the ___ depends on the ___."
- How much does y change each time x goes up by one?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/9-1-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it